Bayesian Hierarchical Logistic Regression for 30-Day Hospital Readmission

The Role of Medical Specialty Among Diabetic Patients

Selina You, Zhouhan Qian

Introduction to Applied Bayesian Analysis | March 2026

Part 1

Motivation & Research Question

Why care about hospital readmissions?

$26B+

Annual cost of unplanned readmissions in the US

- Under the Centers for Medicare & Medicaid Services Hospital Readmissions Reduction Program, hospitals face financial penalties for excess readmissions
- As a result, the 30-day readmission rate has become a central quality metric tied to Medicare reimbursement
- Among all patient populations, diabetes Mellitus patients represent a particularly high-risk population, characterized by complex comorbidities, frequent hospitalizations, and care spanning multiple specialties

How much specialty-level heterogeneity exists in 30-day readmission risk, and can we reliably estimate it despite severe group imbalance?

Why a Bayesian Hierarchical Approach?

Partial Pooling

Small-sample specialties (e.g., PMR, n=288) borrow strength from larger groups through shared hyperparameter τ, yielding more stable estimates than fixed effects or complete pooling.

Coherent Uncertainty

Posterior credible intervals for τ directly quantify uncertainty about between-specialty variation, rather than reducing to a single p-value.

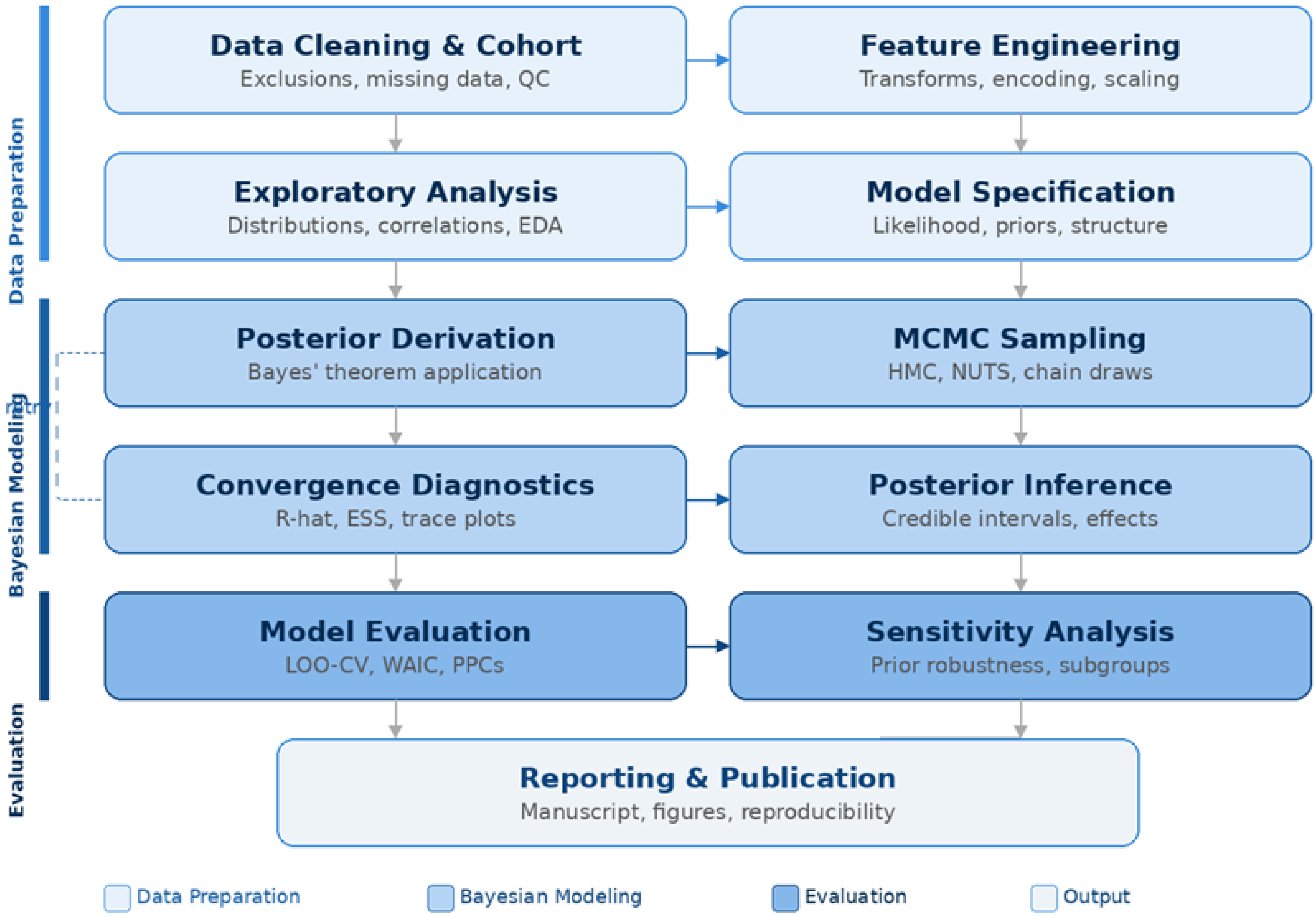
Principled Comparison

DIC and posterior predictive checks provide model assessment tools native to the Bayesian framework.

Part 2

Overall Analysis Framework

Analysis Pipeline Overview



Cohort Construction

Raw Data	**N = 101,766**
Remove duplicates	**−30,248**
Exclude expired/hospice	**−1,545**
Exclude unknown and very small specialty	**−32,008**
Analytic Cohort	**N = 37,964**
Drop or handling High-Missing Vars	
Feature Engineering	**10 continuous + 33 binary/categorical**
Specialty Regrouping	**73 raw specialties -> 19 groups**

Dataset: UCI Diabetes 130 Hospitals dataset
Time period: 1999–2008
Population: Patients with Diabetes Mellitus

37,964
encounters

19
specialty groups (J)

9.60%
readmission rate

43
covariates (K)

Data Overview

Figure 1. Distribution of Observations Across Specialty Groups After Regrouping

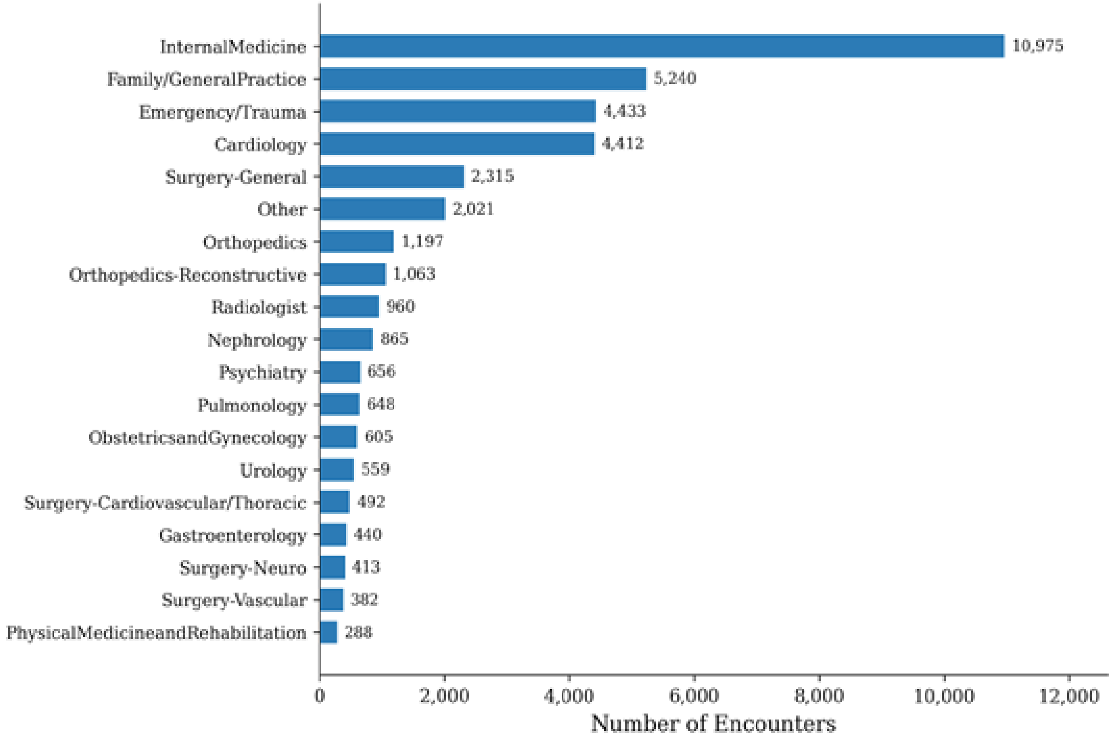
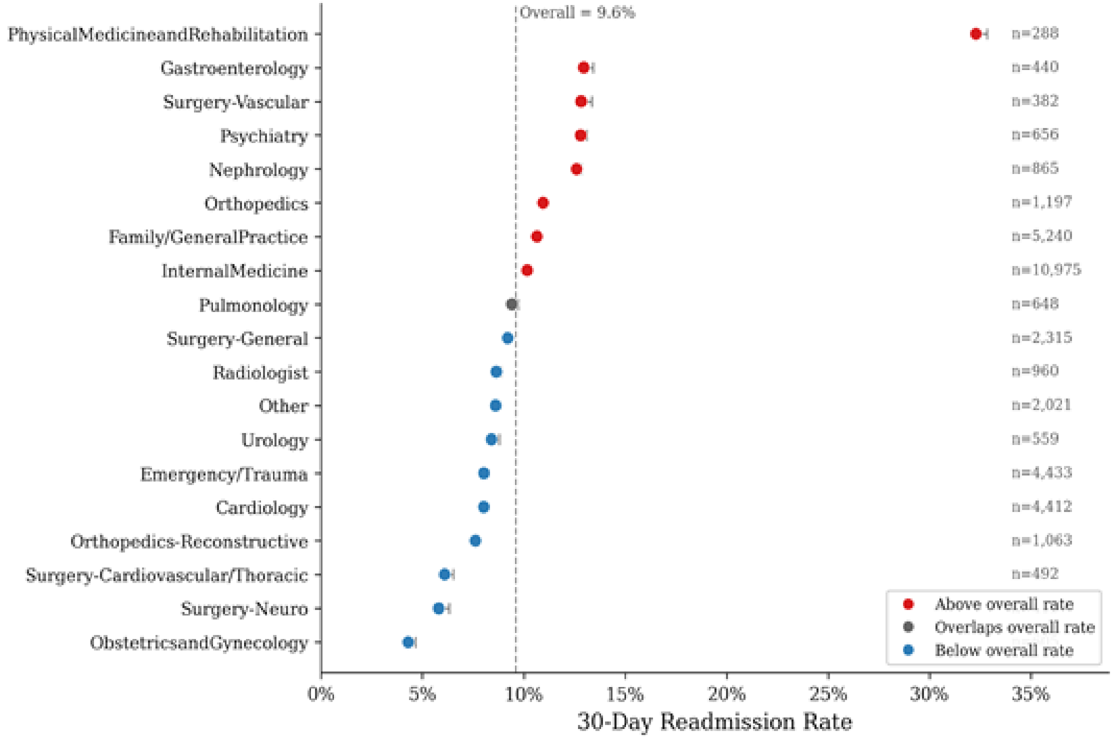


Figure 3. Observed 30-Day Readmission Rates by Specialty Group (with 95% Wilson CI)



Key observations driving this question: Raw readmission rates vary widely across specialties 4.3% to 32.3%, but sample sizes are highly imbalanced

Part 3

Model Structure

Model Specification — Likelihood

$$y_i \mid \alpha, \boldsymbol{\beta}, \mathbf{u} \sim \text{Bernoulli}(\pi_i)$$

$$\eta_i = \alpha + \mathbf{x}_i^\top \boldsymbol{\beta} + u_{j[i]}$$

$$\pi_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}$$

$$\text{Conditional Independence}: \quad y_i \perp y_{i'} \mid \eta_i$$

where:

alpha = global intercept | beta = K=43 fixed effects | u_j = J=19 specialty random intercepts

Prior Specification

$$\alpha \sim \mathcal{N}(0,\ 5^2)$$

Weakly informative on log-odds scale;
covers plausible baseline readmission rates

$$\beta_k \sim \mathcal{N}(0,\ 2.5^2)$$

Allows OR up to exp(5) ~ 150;
prevents extreme inflation without over-constraining

$$u_j \mid \tau \sim \mathcal{N}(0,\ \tau^2)$$

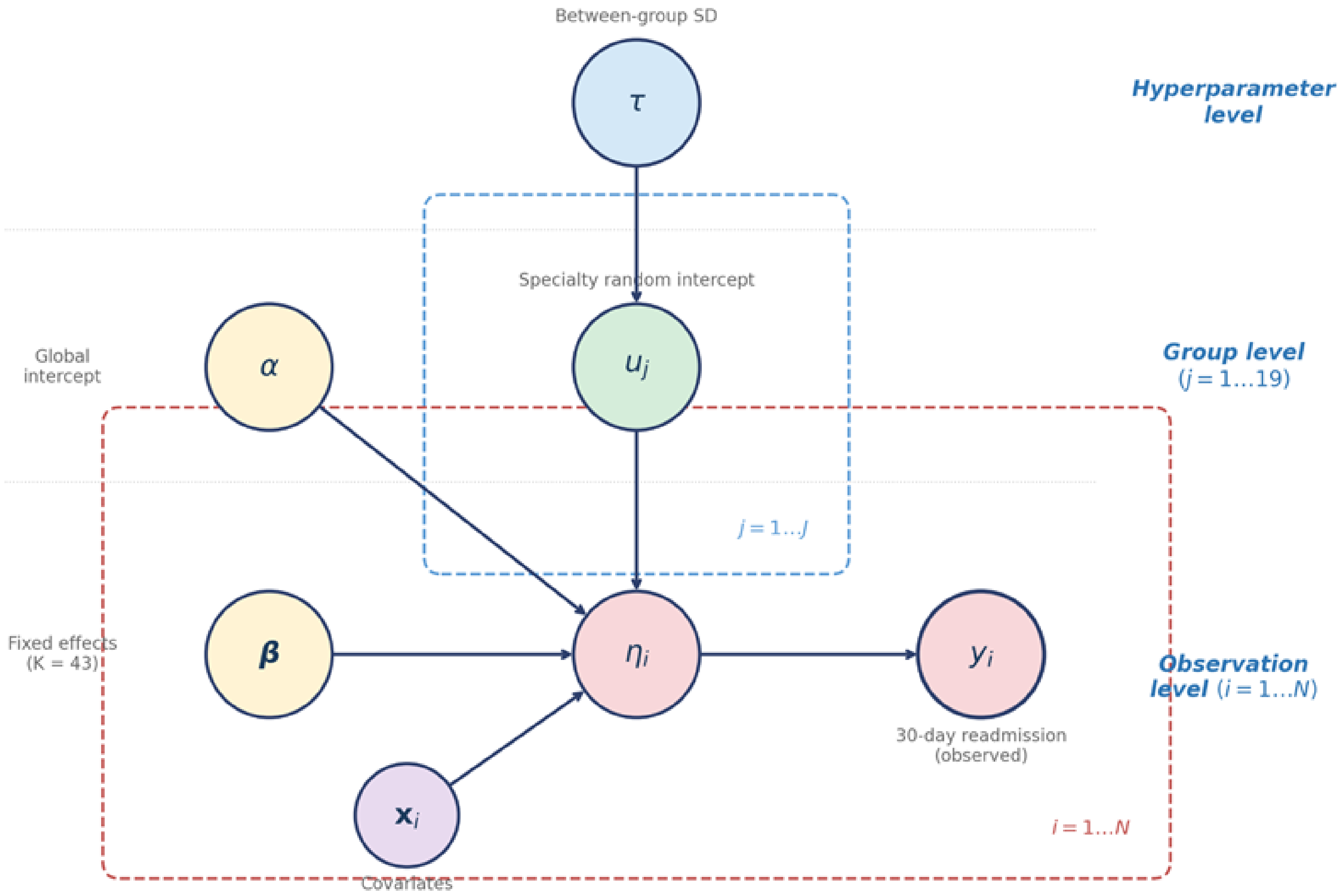
Exchangeable random intercepts;
enables partial pooling across specialties

$$\tau \sim \mathrm{Half\text{–}Normal}(0,\ 1)$$

Hyperprior on between-group SD;
controls degree of shrinkage

The key mechanism: u_j | tau ~ N(0, tau^2) induces partial pooling.
Small groups are shrunk toward the population mean; tau controls shrinkage strength.

Model Hierarchy (DAG)



Model Assumptions

(i) **Conditional independence**
Given parameters, outcomes are independent across encounters: $y_i \perp y_{i'} \mid \eta_i$

(ii) **One observation per patient**
First encounter retained per patient — removes within-patient dependence.

(iii) **Specialty assignment is exogenous**
We do not model the specialty assignment mechanism. Specialty effects should be interpreted as residual heterogeneity, not causal effects.

(iv) **Random intercepts only**
No random slopes — covariate effects are assumed constant across specialties. This keeps the parameter space manageable (64 vs. 64 + 19K).

(v) **Prior independence**
$\alpha \perp \beta \perp \tau$ a priori; u depends on τ only. This simplifies the joint posterior factorization.

Part 4

Key Derivation

Joint Posterior Distribution

$$p(\alpha, \boldsymbol{\beta}, \mathbf{u}, \tau \mid \mathbf{y}) \propto \prod_{i=1}^{N} p(y_i \mid \eta_i) \cdot p(\alpha) \cdot \prod_{k=1}^{K} p(\beta_k) \cdot \prod_{j=1}^{J} p(u_j \mid \tau) \cdot p(\tau)$$

Likelihood	Logistic link, data-dependent
Intercept prior	$\mathcal{N}(0,\ 5^2 = 25)$
Fixed-effect priors	$\mathcal{N}(0,\ 2.5^2 = 6.25)$ each
Hierarchical prior	$u_j \mid \tau \sim \mathcal{N}(0,\ \tau^2)$
Hyperprior	$\text{Half-Normal}(0,\ 1)$

The posterior is non-conjugate due to the logistic likelihood.
No closed-form solution exists → This motivates MCMC sampling.

Log-Posterior Decomposition

$$\ell(\alpha, \boldsymbol{\beta}, \boldsymbol{u}, \tau) = \underbrace{\sum_{i=1}^{N} [y_i \eta_i - \log(1 + e^{\eta_i})]}_{\text{(I) log-likelihood}} - \underbrace{\frac{\alpha^2}{2\sigma_\alpha^2}}_{\text{(II)}} - \underbrace{\frac{1}{2s_\beta^2} \sum_{k=1}^{K} \beta_k^2}_{\text{(III)}}$$

$$\underbrace{- J \log \tau + \frac{1}{2\tau^2} \sum_{j=1}^{J} u_j^2}_{\text{(IV)}} - \underbrace{\frac{\tau^2}{2s_\tau^2}}_{\text{(V)}} + \text{const.}$$

(I) Log-likelihood	Data-dependent: logistic log-likelihood
(II) Prior on alpha	Gaussian regularization on intercept
(III) Prior on beta	Independent Gaussian on each fixed effect
(IV) Prior on u | tau	Hierarchical: tau controls shrinkage; tau only here, NOT in (I)
(V) Prior on tau	Half-Normal hyperprior on group-level SD

Key observations: tau appears ONLY in terms (IV) and (V), not in the likelihood.
tau is informed by data only indirectly through posterior draws of u_j
the tau MCMC update does NOT require likelihood evaluation --> O(J) cost.

Full Conditional Distributions

For each parameter block, collect terms from the log-posterior $\ell(\alpha, \beta, u, \tau)$ that depend on the parameter being updated:

Block 1: α

$$h(\alpha) = \sum_{i=1}^{N}[y_i\eta_i - \log(1+e^{\eta_i})] - \frac{\alpha^2}{2\sigma_\alpha^2}$$

Block 2: β_k (k = 1..K)

$$h(\beta_k) = \sum_{i=1}^{N}[y_i\eta_i - \log(1+e^{\eta_i})] - \frac{\beta_k^2}{2s_\beta^2}$$

Block 3: u_j (j = 1..J)

$$h(u_j) = \sum_{i\,:\,j[i]=j}[y_i\eta_i - \log(1+e^{\eta_i})] - \frac{u_j^2}{2\tau^2}$$

Block 4: ϕ = log τ

$$h(\phi) = -J\phi - \tfrac{1}{2}e^{-2\phi}\sum_{j=1}^{J}u_j^2 - \frac{e^{2\phi}}{2s_\tau^2} + \phi$$

Key observations:

Blocks 1–3 share the same likelihood terms — they differ only in which prior penalty is added.

Block 3 (u_j) sums only over group-j observations — this is a major computational saving.

Block 4 (ϕ) does NOT involve the likelihood at all. Data inform τ only indirectly through the posterior draws of u_j.

None of these conditionals are available in closed form - Metropolis-Hastings required for every block.

Posterior Propriety & Identifiability

Posterior Propriety

Proposition. The posterior $p(\alpha, \beta, u, \tau \mid y)$ is proper (integrates to a finite value).

Proof: (i) Bernoulli likelihood is bounded: $0 \le p(y \mid \cdot) \le 1$; (ii) All priors are proper distributions.

$\Longrightarrow$ **Joint integrand bounded above by $\prod$(proper priors) → integrates to 1.**

Identifiability

Problem: α and u_j are weakly non-identifiable. Shifting $\alpha \to \alpha+c$ and $u_j \to u_j-c$ preserves η_i:

$$\eta_i^* = (\alpha + c) + \mathbf{x}_i^\top \boldsymbol{\beta} + (u_{j[i]} - c) = \eta_i$$

This creates a ridge in the posterior — α and mean(u) are perfectly negatively correlated.

Resolution:

1. Priors: $p(\alpha)$ and $p(u_j|\tau)$ both centered at 0 → break the shift symmetry.

2. **Computation: Block 5 (joint translation move) explores along this ridge → improves mixing for α.**

Reparameterization: ϕ = log(τ) — Derivation

Why reparameterize?

$\tau > 0$ constraint makes random-walk MH inefficient near boundary (proposals $\tau^* < 0$ wasted). $\phi = \log(\tau)$ maps to unconstrained $\mathbb{R}$.

Step 1. Change of variable: $\tau = e^{\phi}$

Step 2. Jacobian: $\left|\frac{d\tau}{d\phi}\right| = e^{\phi} \implies \log\left|\frac{d\tau}{d\phi}\right| = +\phi$

Step 3. Substitute into terms (IV) + (V) of the log-posterior:

$$(\mathrm{IV}): \quad -J\log\tau - \frac{1}{2\tau^2}\sum_{j=1}^{J} u_j^2 \quad = -J\phi - \frac{1}{2}e^{-2\phi}\sum_{j=1}^{J} u_j^2$$

$$(\mathrm{V}): \quad -\frac{\tau^2}{2s_\tau^2} = -\frac{e^{2\phi}}{2s_\tau^2}$$

Step 4. Combine (IV) + (V) + Jacobian:

$$h(\phi) = -J\phi - \frac{1}{2}e^{-2\phi}\sum_{j=1}^{J} u_j^2 - \frac{e^{2\phi}}{2s_\tau^2} + \phi$$

Part 5

Sampling Strategy

Metropolis-within-Gibbs: Block Structure

Block 1: alpha
Symmetric random walk; incremental update delta_eta = alpha* - alpha
O(N)

Block 2: beta_k (k = 1..43)
Component-wise update; delta_eta = (beta_k* - beta_k) * X[:,k]
O(N) each

Block 3: u_j (j = 1..19)
Component-wise; likelihood uses only group-j observations (n_j << N)
O(n_j) each

Block 4: log(tau)
Does not involve likelihood; only depends on current u_j draws
O(J)

Block 5: Translation Move
Joint alpha-u update exploiting non-identifiability (see next slide)
O(J) x 10

MwG: Acceptance Probability

Each block uses a symmetric random-walk proposal:

$$\theta^* = \theta^{(t)} + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2_{\text{prop}})$$

Because the proposal is symmetric — $q(\theta^*|\theta) = q(\theta|\theta^*)$ — the Hastings ratio cancels:

$$\alpha_{\text{acc}} = \min\left(1,\ \exp\left[h(\theta^*) - h(\theta^{(t)})\right]\right)$$

where h(·) is the log-target (full conditional) for the block being updated.

Application to each block:

Block 1 (α): $h = \sum_{i=1}^{N}[y_i\eta_i - \log(1+e^{\eta_i})] - \frac{\alpha^2}{2\sigma_\alpha^2}$ **O(N)**

Block 2 (β_k): $h = \sum_{i=1}^{N}[y_i\eta_i - \log(1+e^{\eta_i})] - \frac{\beta_k^2}{2s_\beta^2}$ **O(N) per k — incremental Δη**

Block 3 (u_j): $h = \sum_{i:j[i]=j}[y_i\eta_i - \log(1+e^{\eta_i})] - \frac{u_j^2}{2\tau^2}$ **$O(n_j)$ — group-j only**

Block 4 (ϕ): $h(\phi) = -J\phi - \frac{1}{2}e^{-2\phi}\sum_j u_j^2 - \frac{e^{2\phi}}{2s_\tau^2} + \phi$ **O(J) — no likelihood!**

Block 5: Joint α–u Translation Move

Motivation:

α and u_j are weakly non-identifiable. Individual MH explores the α–u ridge very slowly. The translation move directly proposes along the ridge.

Step 1. Propose:

$$\delta \sim \mathcal{N}(0, \sigma^2_{\text{blk}}), \quad \alpha^* = \alpha + \delta, \quad u_j^* = u_j - \delta \ \forall j$$

Step 2. Show η* = η (likelihood invariance):

$$\eta_i^* = (\alpha + \delta) + \mathbf{x}_i^\top \boldsymbol{\beta} + (u_{j[i]} - \delta) = \eta_i$$

Likelihood cancels exactly! → Acceptance ratio depends only on priors → O(J) cost

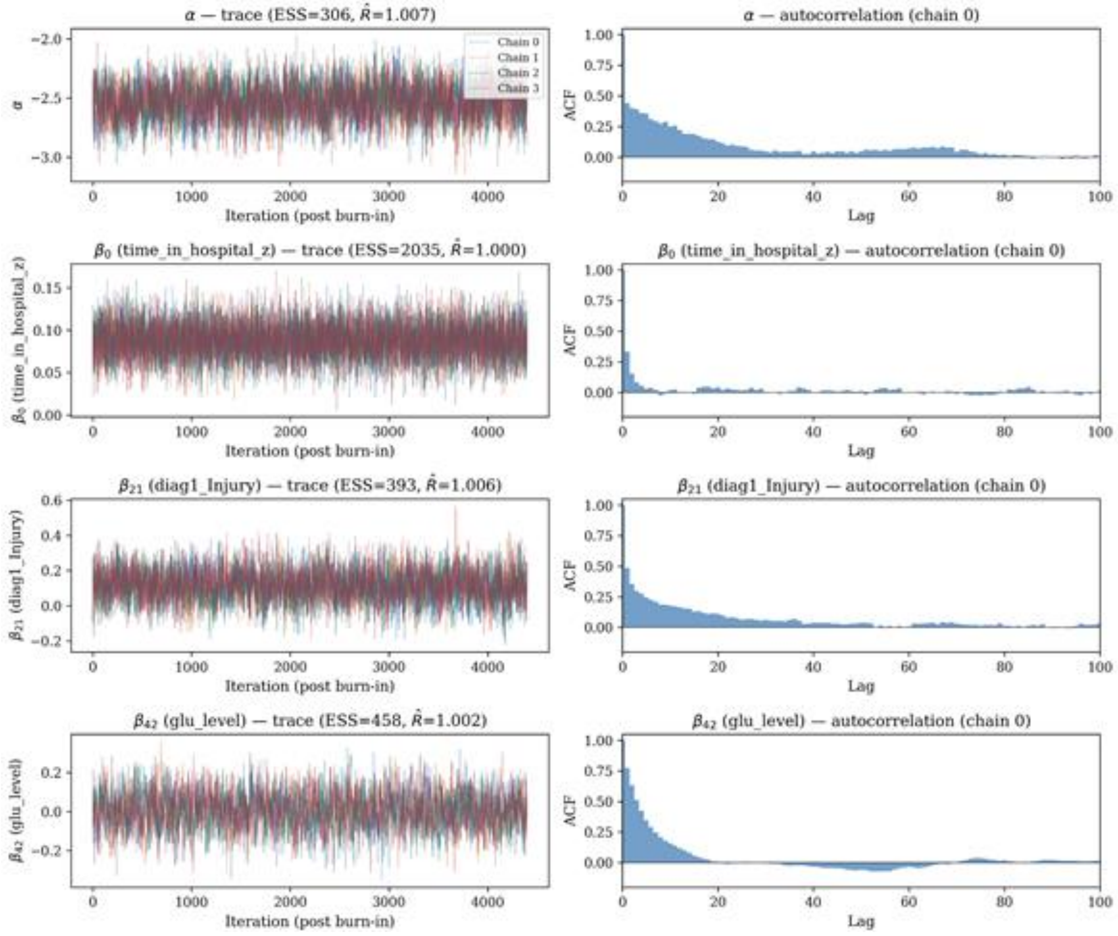
Step 3. Log acceptance ratio (only priors remain):

$$\log R = \left[-\frac{\alpha^{*2}}{2\sigma_\alpha^2} - \sum_j \frac{u_j^{*2}}{2\tau^2} \right] - \left[-\frac{\alpha^2}{2\sigma_\alpha^2} - \sum_j \frac{u_j^2}{2\tau^2} \right]$$

Implementation: Repeated R = 10 times per iteration (cost: 10 × O(19) = O(190), negligible vs O(N) blocks). **Acceptance rate: 0.45.**

Trace Plots — Fixed Effects & Intercept

4 chains | 30,000 iter | Burn-in: 8,000 | Thin by 5 → 4,400 saved / chain | Adaptive per-coordinate tuning (target: 20%–50%)



Parameters shown:

α (intercept, ESS=306),

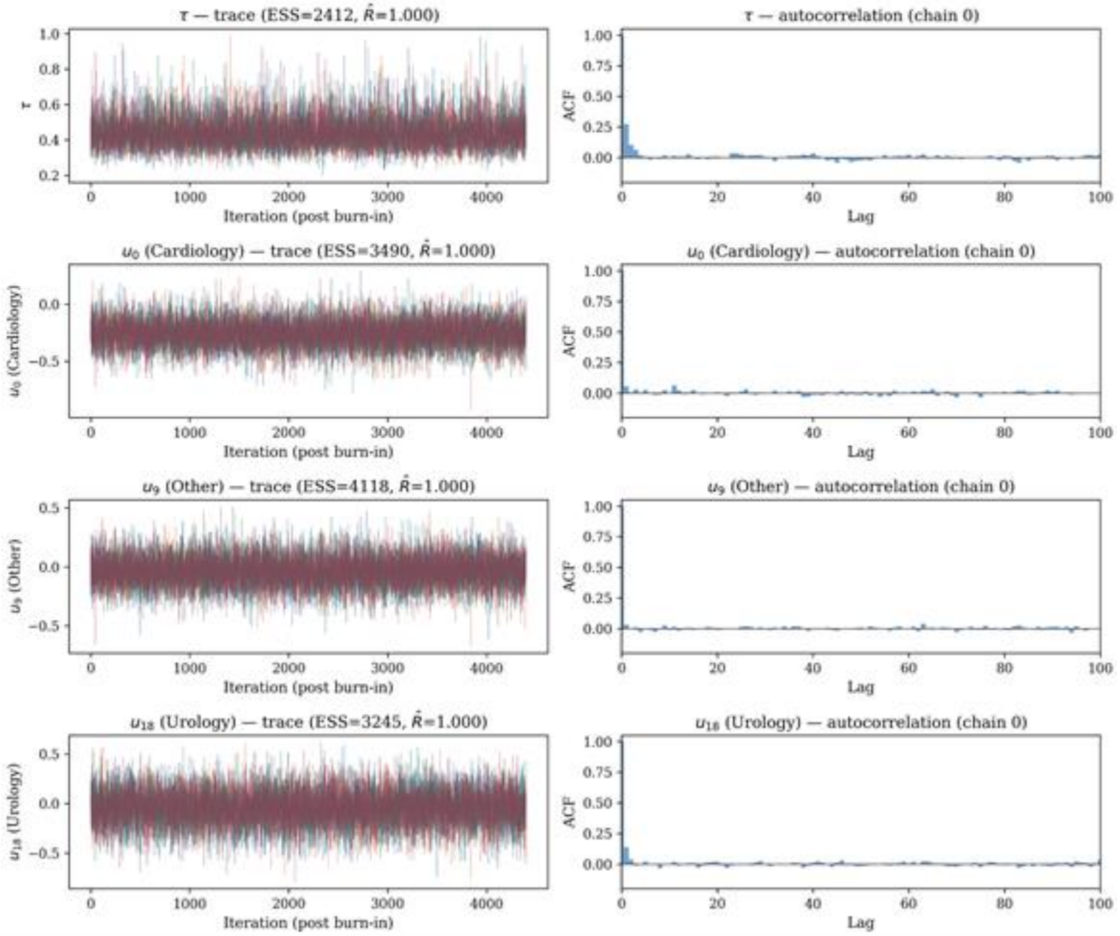
β_0 (time_in_hospital, ESS=2035),

β_{21} (diag1_Injury, ESS=393),

β_{42} (glu_level, ESS=458)

Trace Plots — Hyperparameter τ & Random Effects

All 4 chains overlap and mix well | Autocorrelation decays to near zero within 20–50 lags for most parameters



Parameters shown:

- τ (ESS=2412, $\hat{R}$=1.000),
- u_0 Cardiology (ESS=3490),
- u_9 Other (ESS=4118),
- u_{18} Urology (ESS=3245)

Effective Sample Size & Split-$\hat{R}$

Parameter Block	Min ESS	Median ESS	Max $\hat{R}$	All $\hat{R} < 1.05$?
α (intercept)	245	306	1.007	✓ Yes
β (43 fixed effects)	118	866	1.013	✓ Yes
τ (between-group SD)	2,286	2,412	1.000	✓ Yes
u (19 random intercepts)	1,593	3,490	1.001	✓ Yes

Interpretation

All 64 parameters satisfy $\hat{R} < 1.05$ (max = 1.013 for β_{28}), well below the conventional threshold.

All parameters have ESS > 100; most exceed 500. The hyperparameter τ achieves ESS ≈ 2,412 with $\hat{R} = 1.000$, indicating excellent mixing for the key parameter of interest.

α has the lowest ESS (~306) due to the α–u ridge correlation, partially mitigated by the Block 5 translation move.

Part 6

Inferential Findings

Fixed Effects — Key Predictors

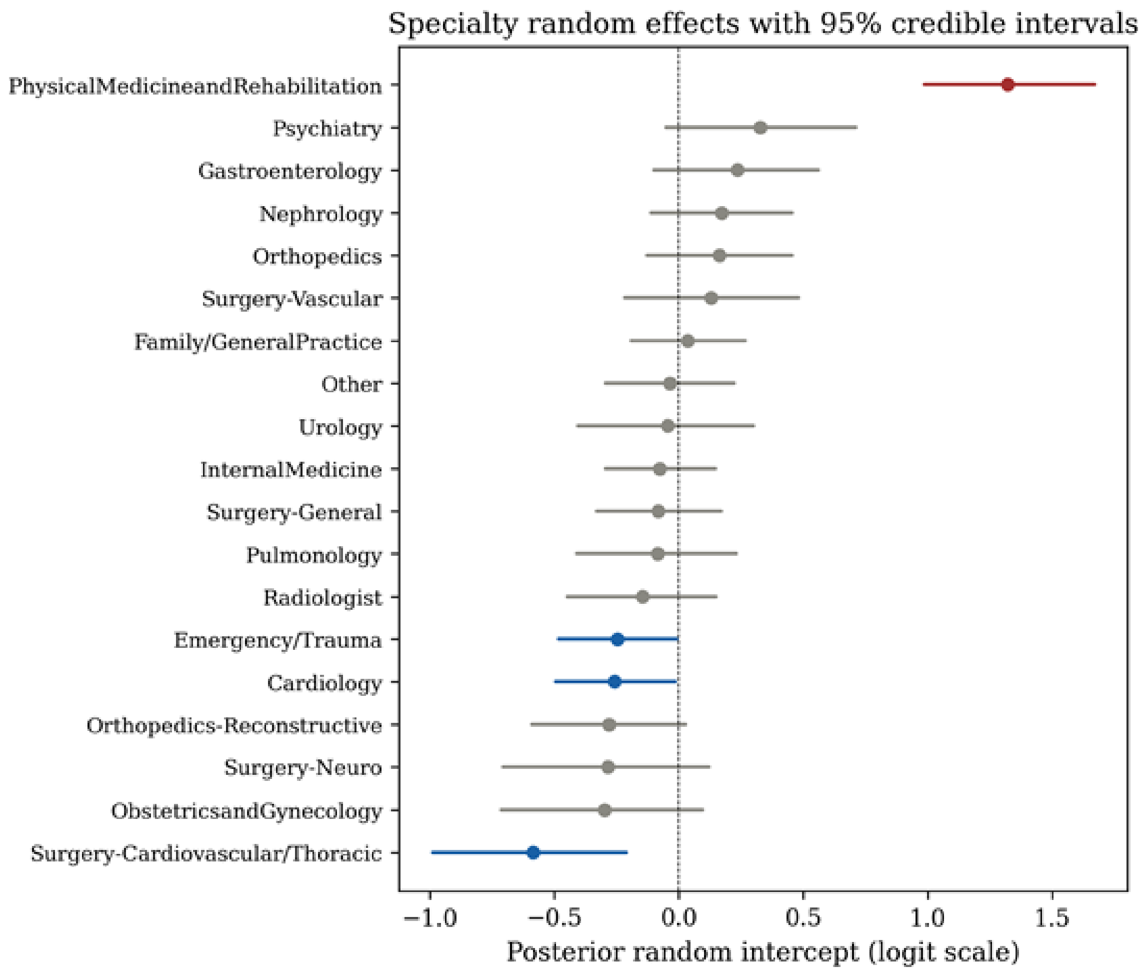
Effects on log-odds scale

Variable	Post. Mean	95% CrI	OR	OR 95% CrI
No. prior inpatient (z)	0.201	[0.175, 0.227]	**1.22**	[1.19, 1.25]
Comorbid neoplasms	0.415	[0.198, 0.631]	**1.51**	[1.22, 1.88]
A1C measured	0.352	[0.154, 0.552]	**1.42**	[1.17, 1.74]
Diabetes medication	0.168	[0.057, 0.277]	**1.18**	[1.06, 1.32]
Age (z)	0.165	[0.123, 0.208]	**1.18**	[1.13, 1.23]
Diag: Neoplasms	−0.507	[−0.792, −0.217]	**0.60**	[0.45, 0.81]
Diag: Respiratory	−0.360	[−0.519, −0.200]	**0.70**	[0.59, 0.82]

Interpretation

Prior inpatient visits strongest predictor (OR=1.22/SD). Comorbid neoplasms highest OR (1.51). A1C measurement reflects severity, not causal testing effect. Sex, race, insulin not significant after adjustment.

Specialty Random Effects & Shrinkage



represents the deviation from the population-average log-odds

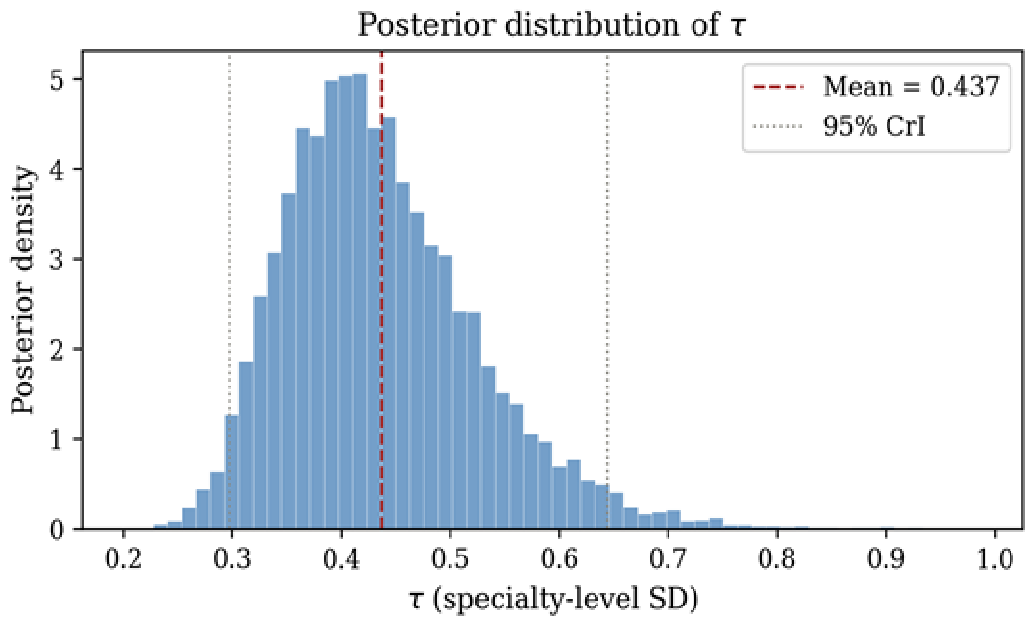
Highest risk Physical Med & Rehab

Lowest risk Pediatrics, OB/Gyn

Shrinkage - Small-sample groups pulled toward the population mean

Key insight - Partial pooling stabilizes estimates for rare specialties

Heterogeneity Scale tau — Core Finding



$\hat{\tau} = 0.437$

95% CrI [0.298, 0.644]

Interpretation

Posterior mass clearly away from zero, which means strong evidence of between-specialty heterogeneity.

Specialty heterogeneity exists and is reliably estimated, but patient-level covariates remain the primary driver of readmission

Model Comparison

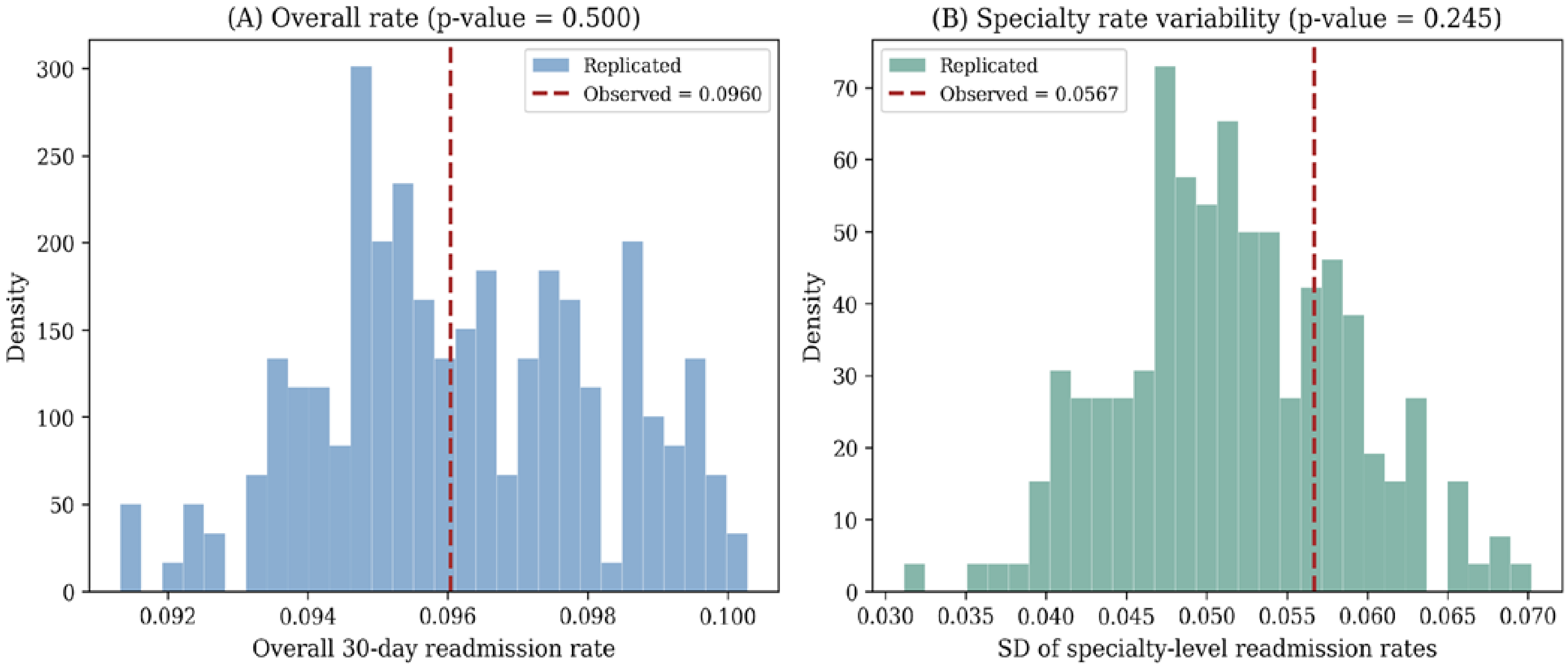
Model	AUC	Brier	Log-Loss	Cal. Slope
No specialty	0.629	0.0852	0.307	1.003
Fixed effects	0.641	0.0846	0.305	0.997
Bayesian hierarchical	**0.641**	**0.0846**	**0.305**	**1.024**

DIC = 23,254 | p_D = 60.5 (< 64 nominal) confirms shrinkage from hierarchical prior

$DIC = \bar{D} + p_D$ effective number of parameters

Take the average of the deviation of the posterior samples by Comorbid neoplasms

Posterior Predictive Check



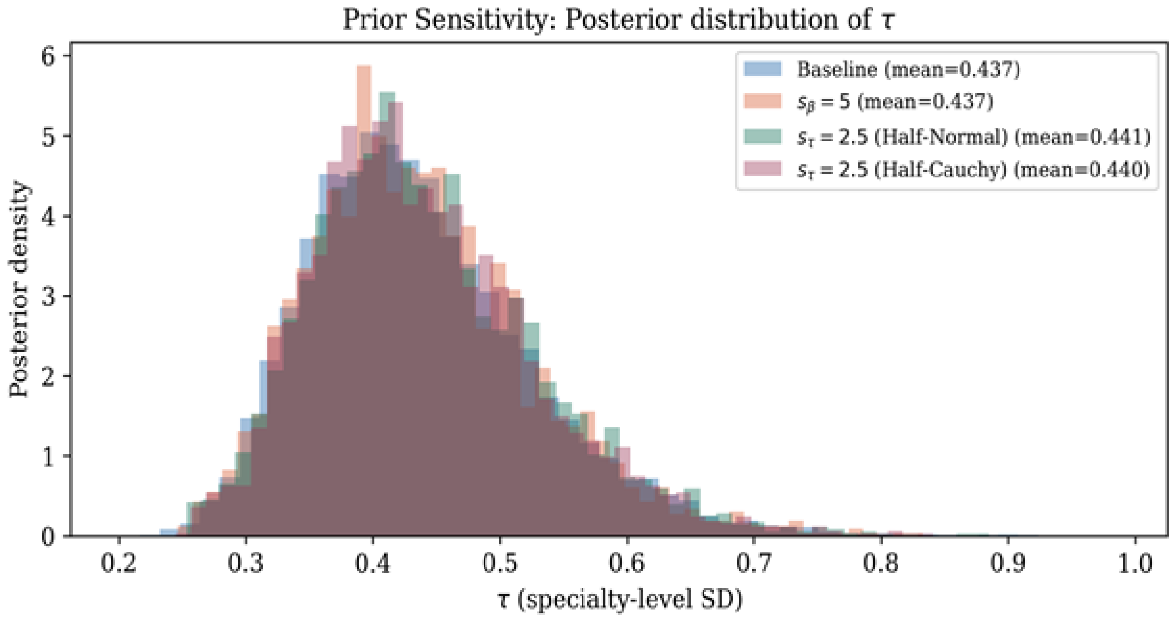
Part 7

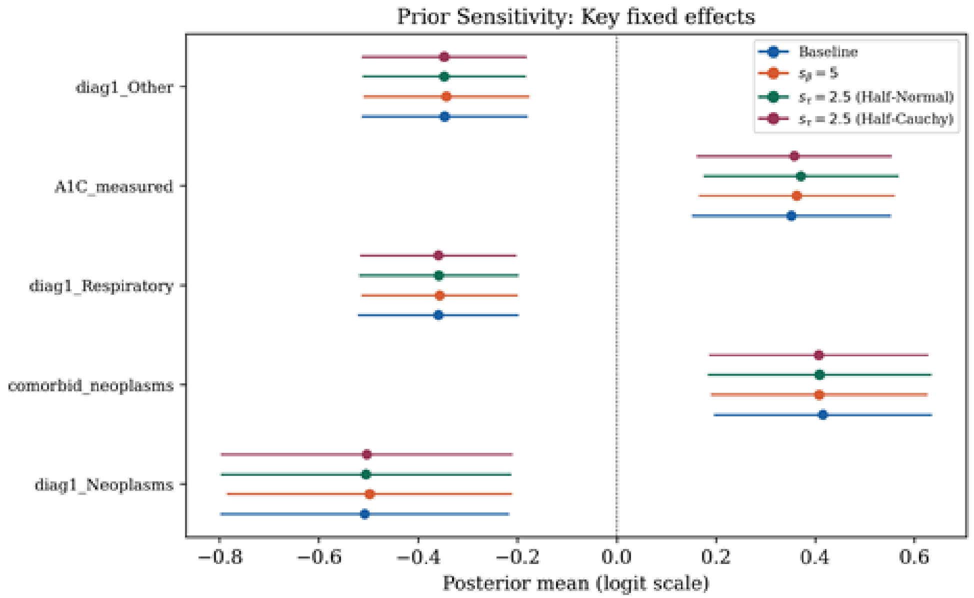
Sensitivity & Robustness Analysis

Sensitivity Analysis — Prior Robustness

Alternative Prior Configurations

Baseline	s_beta = 2.5, s_tau = 1.0 (Half-Normal)
A1	s_beta = 5.0 (fixed-effect prior standard deviation doubled)
B1	s_tau = 2.5 (widen the tau hyperprior to Half-Normal, wider)
B2	s_tau = 2.5 (switch the tau hyperprior to Half-Cauchy, heavier tails)





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Conclusion

tau range across configs: < 0.05

Beta variation: < 0.02 for top predictors

Specialty ranking: Spearman rho > 0.95

Posterior is likelihood-dominated for key parameters. Results are robust to prior choice.

Part 8

Conclusion

Main Findings

Specialty matters **τ = 0.437 (CrI [0.298, 0.644]**)

meaningful heterogeneity beyond patient covariates

Medical complexity drives readmission

Prior inpatient visits, neoplasms, A1C, age, diagnoses are strongest predictors

Model adequacy

PPC confirms model captures both mean and variance structure
DIC favors hierarchical over no-specialty model

Partial pooling

Hierarchical model stabilizes estimates for small specialties via shrinkage toward the population mean

Limitations & Future Work

Limitations

Data vintage	Dataset from 1999-2008, ICD-9 has since transitioned to ICD-10
Selection bias	Excluded 45.7% of encounters with unknown specialty, which could introduce selection bias if the missing-specialty mechanism is non-random.
Model scope	assumes specialty effect is constant across patient profiles

Conclusion

A Bayesian hierarchical logistic regression provides a principled framework for modeling 30-day readmission, capturing both patient-level effects and specialty-level heterogeneity. Compared to fixed-effects models, it enables partial pooling for stable estimates in small specialties, directly quantifies between-specialty variance, and delivers coherent uncertainty. Results are robust to prior choices based on sensitivity analysis.